

Vineel Nadakuditi

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EDUCATION

Rutgers University

B.S. in Computer Science and Data Analytics, GPA: 4.0/4.0, Dean's List

New Brunswick, NJ

September 2022 – December 2026

Related Coursework: Machine Learning Principles, Time Series Modeling, Large-Scale Data Analytics, Data Mining for Business Intelligence, Database Management, Data Structures and Algorithms, Systems Design, Cloud Architecture

Focus Areas: Generative AI, Agent-Based Solutions, Data Pipelines, Data Governance, Predictive Modeling, KPI Definition, Stakeholder Communication

EXPERIENCE

AI Engineering Intern

May 2026 – September 2026

Bristol Myers Squibb

Princeton, NJ

- Designed, developed, and deployed a generative AI solution that automates periodic reviews of validated systems for 20 ITVS users in a regulated environment, reducing FTE hours by 85% and saving \$500,000 per review
- Built data pipelines harmonizing ServiceNow CMDB, DocIT, and the Infinity data lake, improving data quality by resolving 90+ incongruities across systems of record and increasing read/write speeds by 25%
- Designed an evaluation framework measuring LLM output quality across 20 types from Anthropic and OpenAI models, applying prompt engineering and LLM-as-a-judge scoring as a control on generative AI reliability
- Translated business workflows into technical requirements with 6 subject-matter experts, running structured validation cycles that produced 15+ changes and a measured 20% increase in impact

Head Teaching Assistant

January 2025 – May 2026

Rutgers Computer Science Department

New Brunswick, NJ

- Led 800+ students as head TA, communicating technical concepts — data pipelines, database administration, design logic, and error handling — clearly to non-technical audiences
- Analyzed 300+ student responses using AI-driven sentiment analysis and categorization to identify curriculum pain points, then built an AI chatbot for Professor Lau addressing the top findings
- Delivered 8 project demos to 120+ students defining deliverables, baseline expectations, and environment setup, and held biweekly office hours supporting continued learning

PROJECTS

Autonomous Document Review Agent | Python, LangGraph, FastAPI, Postgres, AWS

August 2026

- Architected an agent-based AI pipeline (LangGraph, FastAPI, PostgreSQL) performing concurrent LLM extraction into structured schemas, with Postgres-backed checkpointing for resumable, auditable processing
- Designed a normalized data model with clear taxonomies separating quantitative and qualitative outputs, enabling constant-time trend and comparison queries for ongoing performance measurement
- Implemented data quality controls around LLM output that caught a unit-mislabeling defect prompt engineering could not eliminate, and intercepted a flawed fix before it corrupted production data

Enterprise MCP Server | Python, TypeScript, MCP SDKs, OAuth 2.0, pytest

August 2026

- Designed and published an open-source MCP server exposing enterprise ticketing and CMDB systems as governed, callable tools for LLM agents, implemented across Python and TypeScript SDKs
- Applied production controls for security and scalability — OAuth/service-account authentication, rate limiting, retry with backoff, structured error semantics — validated by an integration test suite and documented with a written spec

Rutgers Health Hack Winning Submission | PyTorch, Pandas, Swift, Firebase, React

November 2024

- Led ML model development in PyTorch for a health tracking app, conducting 6 expert interviews and journey mapping with 10+ heavy users to define hypotheses and target metrics
- Built a RAG pipeline analyzing trends across 5,000+ rows of time-series data and returning natural-language recommendations, earning 1st place and \$3,000 in prize money

TECHNICAL SKILLS

Technical Skills: Python, SQL, PyTorch, AWS, Docker, Terraform, LLM Integration, Prompt Engineering, Generative AI, Data Pipelines, Data Governance, Data Quality, Feature Engineering, Statistical Analysis, Dashboard Construction, Excel, Git, PostgreSQL, Linux, Windows

Certifications: AWS Cloud Practitioner (950/1000), Data Manipulation in SQL, Object-Oriented Programming in Python

Interests: Hiking, Swimming, Golf, Fishing, Chess, Sudoku, New York Times Mini Games